Miniaturized Broadband Antenna Loaded With Frequency Selective Surface for Corona Discharge Detection

Pan Guo, Rui Wang, Liling Wang, and Yi Zhao

***Abstract*— Corona discharge is a common phenomenon on high-voltage transmission lines that can lead to insulation damage and power loss. Antenna detection is one of the current standard detection methods, but it suffers from limited sensitivity. To address this issue, this article proposes a frequency selective surface (FSS) array operating in the corona discharge frequency band. The FSS array has 15 × 15 identical cells. Each cell consists of two metal patches. The upper patch is constructed of a twisted cross-shaped cell, while the bottom patch is constructed of a cross cell with patch inductive components. The cell's dimensions are 10 × 10 × 1.6 mm, with $S_{11} < -10$ dB in the corona discharge frequency band. We superimpose the array onto the antenna and use ANSYS HFSS module to simulate how the distance between the antenna and the array affects antenna performance. The results suggest that the optimum antenna performance occurs when the distance is 20 mm. Corona discharge experiments revealed that the amplitude of signals detected by FSS loaded on the detection antenna is 128%, 130%, 132%, and 135% higher than the original antenna at detection ranges of 1, 4, 7, and 10 m, respectively. Moreover, the antenna achieves the farthest detection distance of 13 m.**

***Index Terms*— Broadband antenna, corona discharge, frequency selective surface (FSS), miniaturization.**

I. Introduction

POOR contact between line and insulator fittings, insulator surface dirt, insulator aging, line strand breakage, and other events in overhead lines can result in an uneven distribution of the local electric field, which is vulnerable to the corona discharge phenomenon [1], [2]. Corona discharge not only wastes energy, but also interferes with wireless communication. Thus, prompt and precise identification of the corona discharge phenomena is critical for the safe operation of overhead lines 0. Currently, the most common detection methods include manual visual inspection [4], ultraviolet detection [5], and antenna detection [6]. The antenna detection method is the most popular because of its stability of detection and inexpensive cost.

Previous researchers have designed some antennas for detecting corona discharges. Fan [7] proposed a spiral array antenna with a maximum detection distance of 260 m and the size is 2 × 4 m. Xie [8] proposed a single-axis spiral antenna with a detection range of about 200 m. The antenna is 360 mm in diameter and 2284 mm in height. However, these large antennas can only be used to detect the presence or absence of corona discharges, and cannot determine exactly where they occur. Therefore, researchers have begun to design small antennas that can be carried by unmanned aerial vehicle (UAV) to locate where corona discharges occur. Yin et al. [9] designed a 5th-order Hilbert fractal antenna with dimensions of 192 × 192 × 1.6 mm for UAV carry detection. Xu and Hei [10] combined Hilbert curves and helical curves to design a Hilbert helical antenna with dimensions of 100 × 100 × 1.6 mm, which can detect 5 m corona discharge phenomenon at 5 m. However, the detection range of these antennas is limited, making UAV detection too close to overhead lines, potentially causing damage to the UAV. Consequently, enhancing the detection distance of antennas is a primary focus of current research.

In order to improve the detection performance of the antenna, this work presents the frequency selective surface (FSS) in conjunction with the detecting antenna to increase its detection sensitivity. FSS is a unique artificial electromagnetic material structure composed of periodically arranged cells. This periodic structure has selective reflection or transmission properties for incident electromagnetic waves of different frequencies, allowing effective control of the reflection and transmission properties of electromagnetic waves [11], [12]. Currently, FSS has led to a wide range of applications in radar antenna systems [13], electromagnetic shielding [14], and wave-absorbing materials [15]. Currently, the concept of FSS application superimposes the FSS with the detection antenna to improve the antenna's capacity to capture electromagnetic waves through the refraction and reflection of electromagnetic waves between the two layers. Ranga et al. [16] proposed that superimposing a double-layer FSS on an ultra-wideband slit antenna could improve the antenna's gain and operating bandwidth, and improve the antenna's ability of capturing electromagnetic waves. This work proved that superimposing FSS on the antenna can improve the antenna's ability to capture electromagnetic waves. In addition, compared to other

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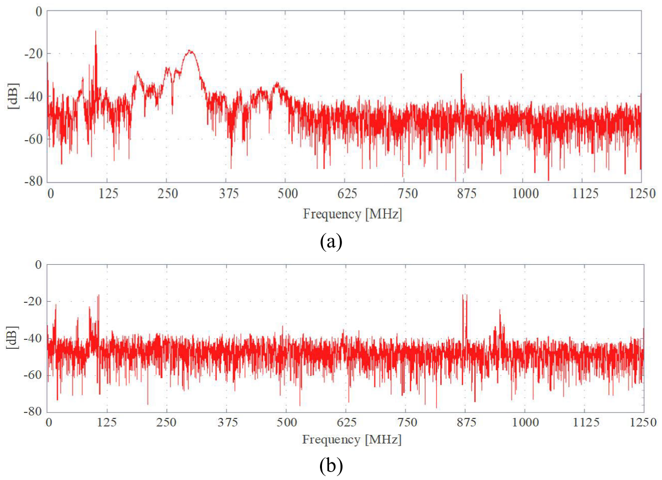

Fig. 1. (a) Spectrum of electromagnetic waves released by corona discharge. (b) Noise spectrum near overhead line [17].

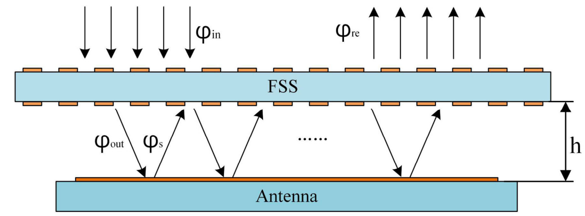

Fig. 2. Operating principle of FSS array on stacked antennas.

devices, FSS is relatively simple to design and its performance can be optimized by adjusting parameters such as cell structure, arrangement and period. At the same time, FSS is also relatively easy to manufacture, and can be fabricated using the traditional PCB process, which is inexpensive and easy to mass produce. So we choose to design the FSS structure.

This article is organized into the following sections. In Section II, we propose a small broadband FSS operating at 100–500 MHz and provide its detailed parameters. In Section III, we simulate the detection antenna with a superimposed FSS and compare it to the antenna without FSS to analyze the effect of the FSS on the antenna's ability to capture electromagnetic waves. In Section IV, we conduct measurements on the detection antenna with a superimposed FSS and establish a corona discharge platform for comparison with the antenna without FSS to assess the ability to capture the corona discharge signal.

II. Design of Frequency Selection Surfaces

A. Design Requirements

According to previous research [17], [18], the spectrum of electromagnetic waves released by corona discharge and noise near the overhead line is shown in Fig. 1. In the range of 100–500 MHz, the energy of the signal is more obvious, and the attenuation is great after 500 MHz, so the signal above 500 MHz is not measured. The frequency band before 100 MHz overlaps with the frequency band of FM radio and other signals, so in order to avoid the measurement being interfered by FM radio signals, the frequency band below 100 MHz is also not measured. So we set it to 100–500 MHz. Therefore, we need to design a frequency selection surface with a resonance frequency of 100–500 MHz.

The principle of the FSS superimposed on the antenna to improve the sensitivity is shown in Fig. 2, and the specific process is as follows.

1) When the incident electromagnetic wave φ_{in} passes through the FSS structure, it transforms into the outgoing electromagnetic wave φ_{out}.
2) The outgoing electromagnetic wave φ_{out} serves as the incident wave of the detection antenna. A part of it is captured by the antenna, and the remaining part generates a reflected electromagnetic wave.
3) The reflected electromagnetic wave φ_s is then incident on the backside of the FSS, and after passing through the FSS, it generates another reflected wave that is incident on the surface of the detection antenna. This cyclic operation can be repeated to increase the total amount of electromagnetic waves captured by the detection antenna.

In summary, the FSS should operate between 100 and 500 MHz to detect corona discharges. When the S_{11} of FSS is less than -10 dB, FSS absorbs 90% of the input electromagnetic wave and reflects the rest, which is sufficient to meet FSS's operating requirements.

In addition, grating lobes impact FSS performance. Grating lobes are generated by superimposing the side lobes of the antenna components in the array. Their scatters electromagnetic waves in unanticipated directions away from the initial reflection point, resulting in energy loss and compromising the FSS's transmission performance. The grating lobes have a reduced influence on the working band as the cell size decreases.

In summary, the design requirements of the FSS are as follows: 1) the operating frequency band of FSS should cover 100–500 MHz; 2) FSS's S_{11} should be <-10 dB; and 3) the size of the designed FSS should be small enough to prevent the grating lobes effect.

B. Design of the FSS

In order to construct a suitable FSS cell, we a single-layer FSS structure must first be created. Next, we will determine its equivalent circuit by analyzing the electric field and current of the single-layer FSS structure. Finally, we will identify the metal patch to be added to the backside of the single-layer FSS using the equivalent circuit, which will lead us to the final FSS cell structure and fulfill the design requirements. The design process of the FSS is described in detail below.

1) Design of the Single-Layer FSS: Fig. 3 depicts the gradual evolution of a single-layer FSS cell structure and its corresponding S_{11} curve. Initially, a basic cross cell design, labeled as pattern (1) in Fig. 2 is shown. The metal patch selected is securely bonded to the upper surface of the dielectric substrate, which measures $10 \times 10 \times 1.6$ mm and is made of FR-4 material. The pattern linewidth, w1, is 0.2 mm. We improved the design by bending and extending the four jib ends of the cross pattern at two consecutive 90° angles to

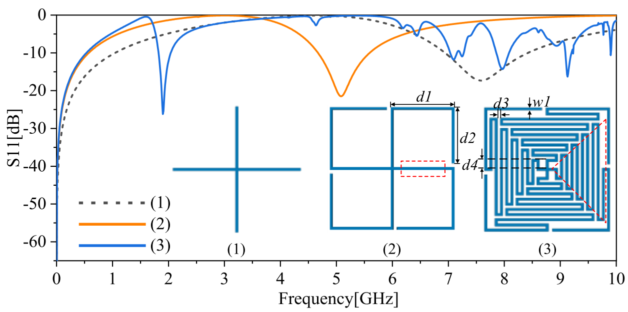

Fig. 3. Conversion process of the signal-layer FSS.

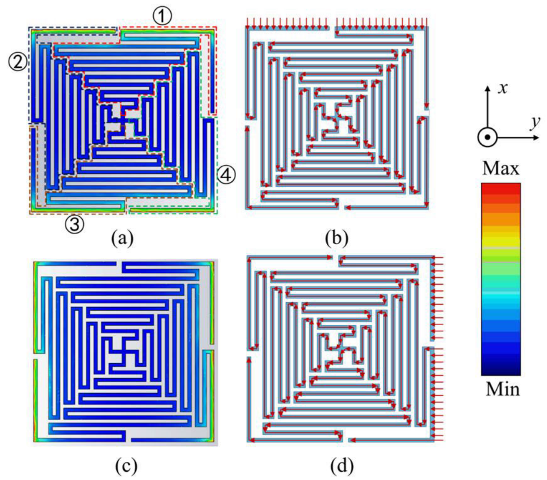

Fig. 4. Electric field component's-direction is in the -x-direction. (a) Electric field distribution. (b) Surface current distribution. Electric field component's-direction is in the -y-direction. (c) Electric field distribution. (d) Surface current distribution.

the right, with extensions of lengths $d_1 = 4.9$ mm and $d_2 = 4.4$ mm, resulting in pattern (2). Subsequently, we modified the red area in pattern (2) into a serpentine curve structure to form the red area in pattern (3), With the spacing between neighboring serpentine structures set at $d_3 = 0.2$ mm. As the structure evolves, the first resonant frequency of the FSS cell, i.e., the frequency at which the resonance first appears, gradually decreases. When the FSS cell structure is optimized to pattern (3), its first resonant frequency stabilizes at 2.133 GHz.

To reduce the FSS's resonant frequency, we added a metal patch on the backside of the dielectric substrate, resulting in a double-layer FSS. According to the design concept of double-layer FSS, we built an equivalent circuit for a single-layer FSS. Establishing the equivalent circuit of FSS can quickly obtain the S_{11} curve of FSS and adjust the circuit parameters based on the design requirements. Modify the structure of FSS according to the circuit to align with the design requirements.

2) Electric Field and Current Distribution: Fig. 4 depicts the electric field and surface current distributions of the FSS's top metal patch at 300 MHz. At other frequencies, the field distribution is similar to that at 300 MHz. Fig. 4(a) shows the single-layer FSS separated into four sections represented by the red, black, yellow, and green boxes, respectively. When the electric field component of the incident electromagnetic wave is in the -x-direction, Fig. 4(a) and (b) show the electric field distribution and surface current distribution of the metal patch respectively. Fig. 4(a) shows that the upper and lower

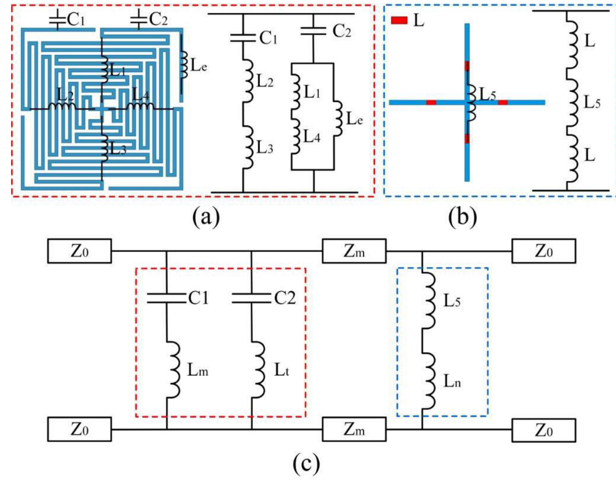

Fig. 5. (a) Equivalent circuit of the upper metal patch. (b) Cross cell and equivalent circuit of the loaded chip inductor. (c) Finalized equivalent circuit.

margins of the patch have larger electric field strength. The surface current in the metal patch flows from region ① to region ②, and from region ④ to region ③. When the electric field component of the incident electromagnetic wave is in the $-y$-direction, these patterns are similar to those observed when the electric field component is in the $-x$-direction as shown in Fig. 4(c) and (d).

3) Construction of Equivalent Circuits: According to the electric field distribution and surface current distribution of the single-layer FSS, the equivalent circuit is shown in Fig. 5(a). In Fig. 5(a), the relatively strong electric field strength of the metal patch layer is equivalent to capacitors C_1 and C_2. Since the pattern as a whole is short-circuited, there are only two equivalent capacitances, C_1 and C_2, and no capacitive effects at other locations. According to the flow direction of surface current, for the surface current generated by the incident electromagnetic wave from the $+x$-direction, there are two directions of flow from region ① to region ② and region ④ to region ③. Therefore, the following equivalent inductances exist in the metal patch, and the four regions of the patch are equivalent to L_1, L_2, L_3, and L_4, respectively, in which the beginning of region ① is equivalent to Le. Subsequently, C_1, L_2, and L_3 are connected in series, and L_1, L_4 are connected in parallel with Le and then in series with C_2. The two parts of the circuit are connected in parallel, and the simplified equivalent circuit is shown as the red box in Fig. 5(c).

Z_0 and Z_m represent the characteristic impedance in free space and dielectric space, respectively. These values can be calculated by the following equations [19]:

$$L = \mu_0 \frac{d}{2\pi} \log\left(\frac{1}{\sin\frac{\pi w}{2d}}\right) \tag{1}$$

$$C = \varepsilon_0 \varepsilon_{\text{eff}} \frac{2D}{\pi} \log\left(\frac{1}{\sin\frac{\pi s}{2D}}\right) \tag{2}$$

$$Z_m = \frac{Z_0}{\sqrt{\varepsilon_{\text{eff}}}}. \tag{3}$$

In (1)–(2), μ_0 represents the free-space permeability with a value of $4\pi \times 10$–7 S/m, d represents the length of the metal strip, w represents the width of the metal strip, ε_0 and

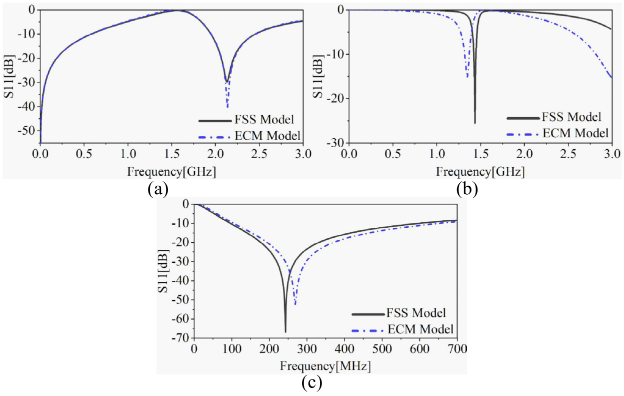

Fig. 6. (a) S_{11} curves of the single-layer FSS and its equivalent circuit. (b) S_{11} curves of the double-layer FSS without chip inductors and its equivalent circuit. (c) S_{11} curve of the double-layer FSS with chip inductors and its equivalent circuit.

ε_{eff} represent the dielectric constants of the free-space and the dielectric substrate, respectively. D represents the length of the two independent metal strips constituting the capacitor, and s represents the spacing between the two independent metal strips. The final equivalent values of the capacitor inductance are calculated as $L_m = 5.262$ nH, $L_t = 46.312$ nH, $C_1 = 173.404$ fF, and $C_2 = 226.823$ fF. In (3), $Z_0 = 377\ \Omega$, and $Z_m = 182\ \Omega$. By substituting the above parameters into the software Advanced Design System 2024 Update 2 for modeling and simulation, the equivalent circuit is obtained. The single-layer FSS's S_{11} is basically the same as the S_{11} curve of its equivalent circuit as shown in Fig. 6(a).

4) Back Structure Design: We define the designed single-layer FSS as the upper patch and the patch added on the back as the bottom patch. Subsequently, we attempt to successfully lower the resonant frequency of the equivalent circuit by adding components on the right side of the Z_m in the equivalent circuit. When the added components are pure inductors, resonant frequency of the whole equivalent circuit reduces. The resonant frequency reduces as the inductance value increasing. In the FSS model construction, when the length of the cross pattern is set to be the same as the edge length of the dielectric substrate, the equivalent circuit presents a purely inductive structure [11]. In view of this, we selected the cross pattern as the bottom patch. The equivalent inductance of cross pattern in this design is calculated as $L_5 = 10.274$ nH according to (1).

Fig. 6(b) shows the S_{11} curves of the FSS and its equivalent circuit with the cross pattern as the bottom patch. The analytical findings show that the cross pattern's equivalent inductance is insufficient to drop the resonance point below 500 MHz. To increase the inductance value, we decide to add patch inductors on each of the four arms of the cross pattern, as shown in Fig. 6(b). The parameters of the patch inductors were determined using the constructed equivalent circuit and the optimization toolbox in the ADS software. The optimization results reveal that setting the patch inductor to 390 nH can meet requirements. This ultimately results in an S_{11} curve of the FSS that closely resembles its equivalent circuit, as shown in Fig. 6(c). The finalized structure of the cell is shown in Fig. 7, where the linewidth $w_2 = 0.4$ mm of the bottom patch.

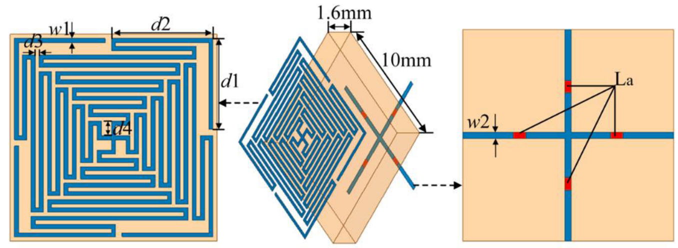

Fig. 7. Overall structure of finalized FSS.

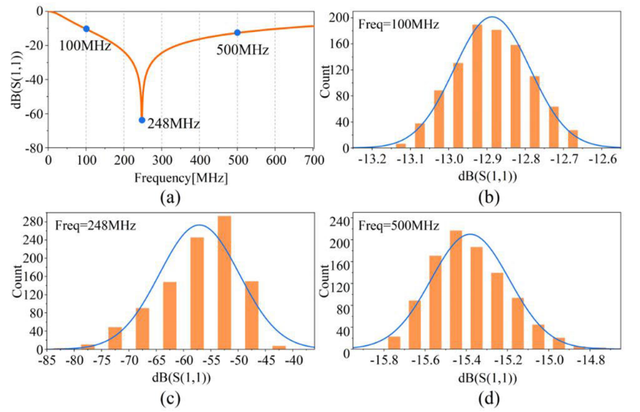

Fig. 8. (a) S_{11} curve of FSS. (b) Histogram of S_{11} distribution at 100 MHz. (c) Histogram of S_{11} distribution at 248 MHz. (d) Histogram of S_{11} distribution at 500 MHz.

Based on the above results, the simulation was conducted using Ansys Electronic Desktop 2022 R2 software to obtain the S_{11} curve of the FSS, as shown in Fig. 6(c). The black line represents the S_{11} curve resulting from the electromagnetic wave incident from the upper metal patch layer, with a resonant frequency of 248 MHz and $S_{11} < -10$ dB within the range of 100–500 MHz. When the electromagnetic wave is incident from the upper metal patch layer, the resonant frequency of the FSS is 248 MHz, and the electrical dimensions are $0.0084\lambda_0\times\ 0.0084\lambda_0\times\ 0.0013\lambda_0$ (where λ_0 represents the wavelength of the resonant frequency). This size can completely eliminate the influence of the grating lobes on the FSS array while meeting the design requirements.

C. Monte Carlo Analysis of Inductance Values

In the previous simulation study, we added patch inductors on the back of the FSS and simulated it at an ideal inductance value. However, considering the tolerance that exists in the patch inductor value (Model MLG1005SR39JT000, nominal 390 nH, $\pm 5\%$ float) in the real engineering application, we conducted Monte Carlo analysis for four chip inductors with FSS.

We set the inductance value to 390 nH and allowed a tolerance variation of $\pm 5\%$ using the Ansys statistical tool. We generated 1000 sets of random numbers to simulate the variation of the inductance value and then assessed the specific effect of the tolerance on the performance of the FSS. The statistical results are shown in Fig. 8, where Fig. 8(a) depicts the S_{11} curves of the FSS. Fig. 8(b)–(d) present histograms of the distribution of the S_{11} parameters at 100, 248, and 500 MHz, respectively. The horizontal coordinates of these histograms

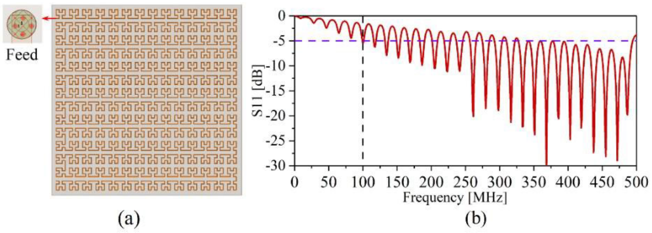

Fig. 9. SHU antenna with corresponding S_{11} curve [20].

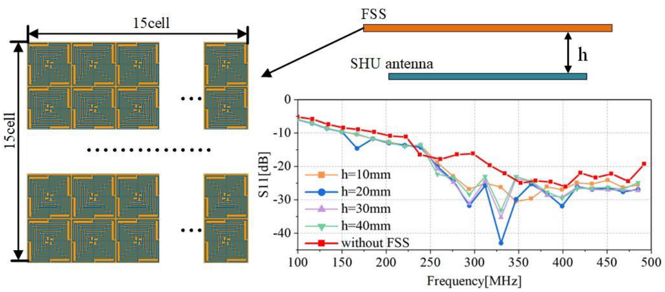

Fig. 10. S_{11} curve of the overall antenna for different heights.

represent the S_{11} parameter values, while the vertical coordinates reflect the frequency of occurrence of each S_{11} parameter value during the Monte Carlo analysis. Among them, Fig. 8(b) shows that at 100 MHz, the S_{11} parameter values are mainly concentrated in the range of −13.2 to −12.6 dB, showing a normal distribution. Fig. 8(c) reveals that at 248 MHz, the S_{11} parameter values are more concentrated in the range of −80 to −40 dB, showing a skewed normal distribution. Finally, Fig. 8(d) shows that at 500 MHz, the S_{11} parameter values are mainly distributed in the range of −15.8 to −14.8 dB, also showing an approximately normal distribution.

Summarizing the above analysis, it can be concluded that while the inductor tolerance has a significant impact on the S_{11} parameter value of the FSS at 248 MHz, the effect is minor at 100 and 500 MHz. This results further confirm that the inductor tolerance influences the S_{11} parameter value at the resonant frequency of the FSS, with minor impact on others frequencies within the FSS's operating band.

III. Simulation and Analysis of Antennas

A. Simulation of FSS-Stacked Antennas

In order to observe the effect of FSS on the detection antenna, we used a previously designed serpentine Hilbert fractal (SHU) antenna as the detection antenna [20]. The antenna dimensions are 130 × 130 × 1.6 mm, and the specific structure and S_{11} curve are shown in Fig. 9. The SHU antenna can detect the corona discharge signals at a distance of 10 m from the source of the corona discharge. The S_{11} curve of the SHU antenna exhibits numerous resonance points, significantly expanding the operating band. To visually represent the working band of the antenna more clearly, we connect the resonance points with a red line, as shown in Fig. 10.

To improve the SHU antenna's performance, we positioned the FSS structure above the antenna at h, as shown in Fig. 10. The dimensions of the SHU antenna are 130 × 130 × 1.6 mm.

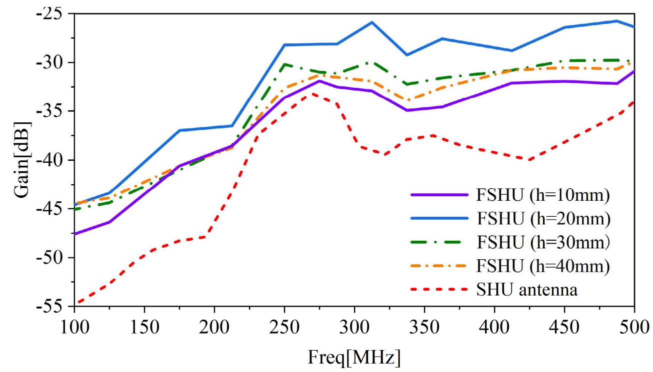

Fig. 11. Gain comparison of antennas.

We set the FSS array as an array of 15 × 15 FSS cells with a size of 150 × 150 × 1.6 mm, which is sufficient to cover the receiving surface of the detection antenna. The distance between the FSS array and the antenna is h. For the sake of simplicity of expression, we refer to the antenna without FSS as the SHU antenna, and the antenna loaded with FSS as the FSHU antenna. When the antenna is too close to the FSS array, it causes a near-field coupling effect, resulting in a degradation of the antenna's performance. As the distance increases, more electromagnetic waves are not reflected back to the antenna but into the air, resulting in fewer electromagnetic waves being received by the antenna, which leads to a decrease in performance [21]. Therefore, the most suitable h needs to be obtained by simulation.

We then set h to 10, 20, 30, and 40 mm to recreate the FSHU antenna's S_{11} curve at varied h values. The generated S_{11} curves have numerous resonance points following the same trend. To demonstrate the distinction, we extracted the resonance points of the S_{11} curves and connected them. The results are depicted in Fig. 10. The yellow, blue, purple, and green lines represent the S_{11} curves of the loaded FSHU antenna under $h = 10$, 20, 30, and 40 mm, respectively, while the red line represents the S_{11} curve of the SHU antenna.

According to Fig. 10, the S_{11} curve of the FSHU antenna is significantly lower than that of the SHU antenna, indicating that the designed FSS can enhance the antenna's ability to receive electromagnetic waves. Specifically, as h increases, the antenna's ability to receive electromagnetic waves peaks at $h = 20$ mm. As h increases beyond 20 mm, the S_{11} curve gradually shifts upward, leading to a decrease in reception ability. Subsequently, we compared the gain of the SHU antenna with FSS superimposed on it with that of the SHU antenna, as shown in Fig. 11. We found that the gain of the antenna is maximized when $h = 20$ mm. Therefore, we decided to set h at 20 mm.

B. Boundary Setting

The boundary setup used in simulation is shown in Fig. 12. The boundary setup for FSS simulation is shown in Fig. 12(a). The upper and lower ports are used as floquet ports, and the boundaries around the perimeter are used as master-slave boundaries. The boundary setup used for the FSHU antenna is simulated as shown in Fig. 12(b). The whole

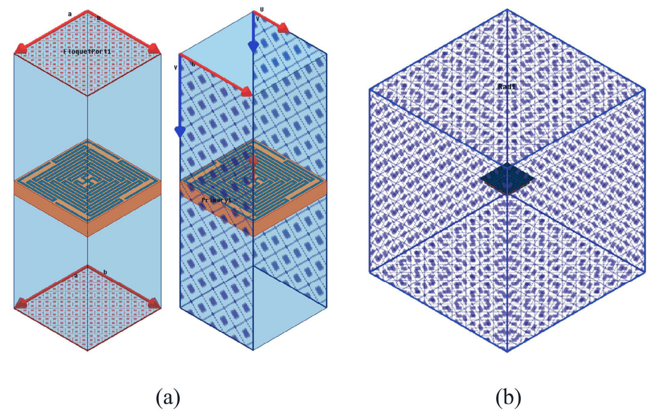

Fig. 12. (a) Boundary setup for FSS. (b) Boundary setup used for the FSHU antenna.

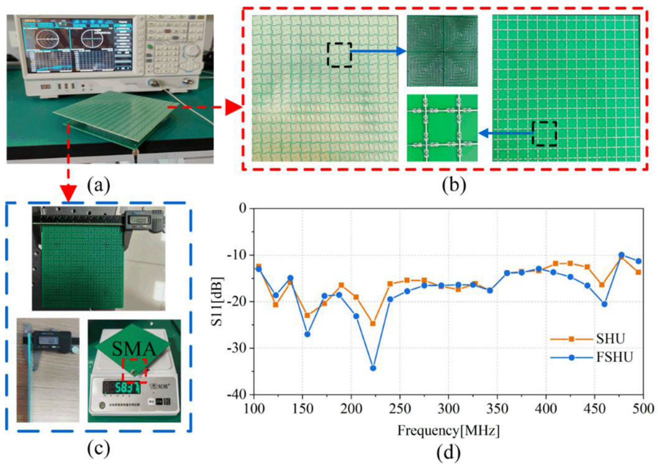

Fig. 13. (a) Antenna connected to network analyzer. (b) Physical frequency selection surface. (c) Physical SHU antenna. (d) Comparison of measured S_{11} curve with simulation.

boundary is used as a radiating boundary, and it is required that the boundary distance from the antenna should be greater than 1/4 resonance wavelength. All the meshes of the simulation are modeled using the CLASSIC model and the mesh density is set to 0.0001 mm.

IV. Experiments

A. Measured Parameters of Antennas

We fabricate the model of the SHU antenna and FSS array using standard circuit board (PCB) technology and machine placement techniques. We stack them together at a spacing of 20 mm, and the finished product is shown in Fig. 13(b) and (c). They are tested separately by a network analyzer, and the resulting S_{11} is shown in Fig. 13(d). The S_{11} curves of the FSHU antenna are lower than those of the SHU antenna in the 100–500 MHz range, with the maximum difference observed in the 200–250 MHz band range.

However, there is a gap between the actual detected S_{11} curves and the S_{11} curves obtained from the simulation due to the following reasons.

1) The SMA interface at the feed point is manually welded, leading to the presence of extra inductance and reactance of the antenna, resulting in differences in the S_{11} curves.

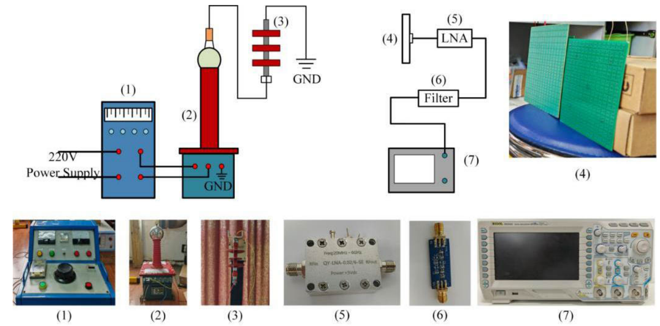

Fig. 14. Experimental platform.

2) The detection environment contains high-frequency electromagnetic noise, such as vehicular electromagnetic interference, which can affect the detection accuracy. Next, we built a corona discharge platform, detected the corona discharge signal using two antennas, and compared the detection gap.

B. Corona Discharge Detection Experiments

The experimental platform used in the corona discharge detection test involving the prototype SHU antenna is shown in Fig. 14. The main components of this experimental platform include the following.

1) YDQB 10 kA/50 kV transformer system (Luzhou Specialized Transformer Company Ltd.) powered by a 220 V ac power supply.
2) Composite insulators with a discharge threshold of 10 kV. The insulators underwent artificial contamination testing following International Electrotechnical Commission (IEC) standard 60 507:2013. A mixture of sodium chloride and kaolin was prepared in a 1:5 ratio, dissolved in deionized water, stirred homogeneously with a brush, coated on the insulators' surface, and allowed to dry naturally.
3) SHU antenna loaded with FSS and unloaded with FSS.
4) Bandpass filters with frequencies ranging from 100 to 500 MHz, were applied to the antenna outputs to eliminate low and excessively high-frequency signals.
5) A low-noise amplifier with a gain of 20 dBm and a frequency range of 20 MHz to 6 GHz was used to amplify the received signal.
6) A RIGOL MSO2202A multichannel oscilloscope was employed to observe, record, and analyze the corona discharge signals captured by the antenna.

Ambient water vapor increases the refraction and attenuation of electromagnetic waves and causes a decrease in the breakdown voltage of air. Additionally, temperature also affects the propagation of electromagnetic waves. Therefore, before experimenting, we tested the temperature and humidity of the environment. The temperature of the experimental environment was 15 °C and the humidity was 59%. We first use two antennas to detect ambient noise while the transformer center terminal was on and the voltage was not yet amplified, and then use two antennas to detect the electromagnetic wave signals released from the amplifier at distances of 1, 4, 7, 10, and 13 m. This distance was chosen because the drone does

TABLE I
AMPLITUDE MAGNITUDE OF THE SIGNAL RECEIVED BY THE ANTENNA AT DIFFERENT DISTANCES

Distance	Maximum amplitude of the SHU acquisition signal (mV)	Maximum amplitude of the FSHU acquisition signal (mV)
1m	400(87.0)	512 (104.13)
4m	56 (13.2)	73 (11.7)
7m	26.8 (5.4)	35.6 (6.3)
10m	12.1 (1.45)	16.4 (1.31)
13m	No value	6.9 (0.65)

not need to be too far away when carrying an antenna for detection. Each antenna was subjected to ten experiments at each distance, with each experiment separated by 1 min to ensure complete release of the electromagnetic wave signal. Their time-domain waveforms were recorded by an oscilloscope. We recorded the amplitude of each set of waveforms and calculated the average amplitude of the time-domain waveforms as well as the variance of each set of waveforms, as shown in Table I. The numerical expression in Table I is in the form of "average amplitude (variance)." We selected a set of time-domain waveforms of the detection results and performed FFT calculations on them. The time-domain and frequency-domain distributions of the corona discharge signal and noise are shown in Fig. 15.

Based on Fig. 15 and Table I, the waveforms of the corona discharge signals detected by the two antennas are essentially the same at detection distances of 1, 4, 7, and 10 m. The maximum amplitudes of the corona discharge signals detected by the FSHU antenna are 128%, 130%, 132%, and 135% higher than those detected by the SHU antenna, respectively. When the detection distance is set to 13 m, the maximum amplitude of the FSHU antenna detection signal in the time domain is 7.5 mV, and the SHU antenna is unable to detect the corona discharge. When setting the detection distance farther away, the oscilloscope can no longer capture the corona discharge signal.

Subsequently, we positioned the two antennas as in Fig. 16, adjusting the placement angle from 0° to 90° in 15° increments. We detected corona discharges at distances of 1, 4, 7, and 10 m, with the results presented in Fig. 17. According to Fig. 17, the amplitude of the corona discharge signal detected by the antenna gradually decreases as the angle decreases. At a distance of 10 m, the FSHU antenna is unable to detect signals at a placement angle of 0°, while the SHU antenna fails to detect signals at both 0° and 15°. And when the distance is set to 13 m, the FSHU antenna can only detect the incident corona discharge signal when positioned at an angle of 90°; it is unable to detect signals at other angles. In summary, the effectiveness of the detection effect decreases when the antenna is oriented to receive oblique incoming electromagnetic waves.

In order to further analyze the capability of the designed antenna to detect corona discharge signals, we calculated the amount of charge capacity captured by the antenna at a distance of 13 m. We fabricated a high frequency Rogowski coil using a nickel-zinc ferrite magnetic ring (anti-interference frequency range: 40–400 MHz) and positioned it on the

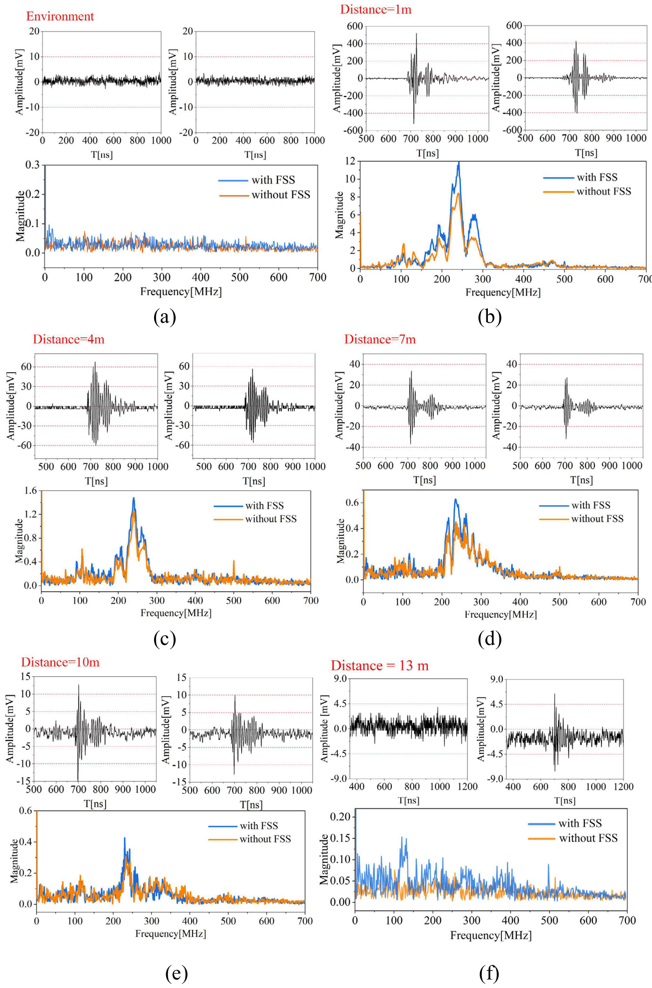

Fig. 15. (a) Environment noise. (b) Detection distance of 1 m. (c) Detection distance of 4 m. (d) Detection distance of 7 m. (e) Detection distance of 10 m. (f) Detection distance of 13 m.

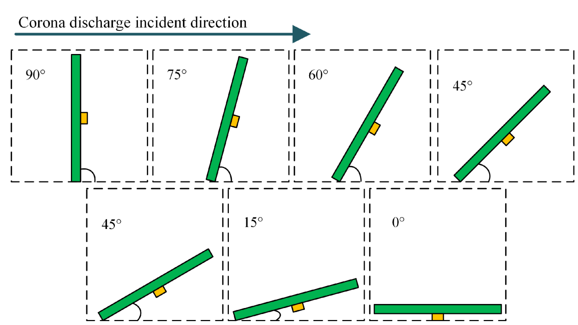

Fig. 16. Schematic of antenna placement angle change.

grounding wire of the insulator. This setup was implemented to measure the amount of charge released during the corona discharge in the insulator experimental circuit.

We adjusted the voltage transformer to 40 kV. When the insulator surface emitted a "hissing" sound, it indicated the occurrence of corona discharge. The generated current passed through the grounding wire, inducing an electromotive force in the Rogowski coil. This electromotive force was then converted into a voltage signal through a 47 Ω resistor. After passing through a 20 dB attenuator, the signal was finally captured by an oscilloscope, as shown in Fig. 18.

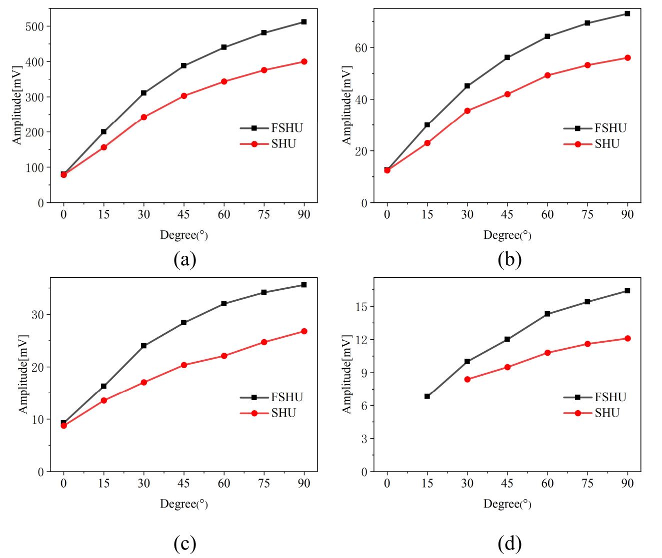

Fig. 17. Average amplitude of the antenna at different distances corresponding to the change in antenna angle. (a) 1 m, (b) 4 m, (c) 7 m, and (d) 10 m.

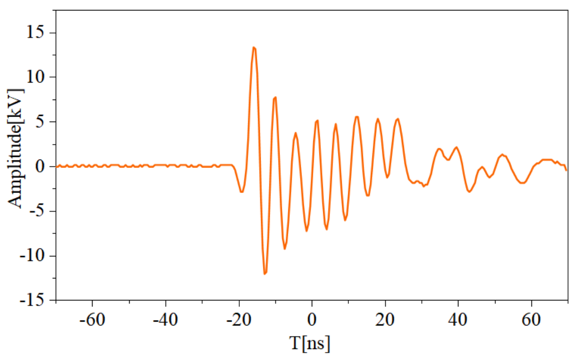

Fig. 18. Voltage time-domain waveform signal detected by the high frequency Rogowski coil when corona discharge occurs in the insulator experimental circuit.

We used the equation [4] in [23] to calculate the charge capacity

$$Q = \frac{1}{R}\int_{t2}^{t1} u(t)dt. \tag{4}$$

In equation [4], R represents the resistance value of the resistor that converts the electromotive force into a voltage signal, which is 47 Ω. The $u(t)$ represents the function corresponding to the voltage waveform. The calculated charge capacity released by the insulator experimental circuit was 0.000302 C. Subsequently, we utilized the equation proposed in [24] to calculate the amount of charge capacity that the designed antenna can capture

$$Q_{\text{ant}} = \frac{Q_{\text{int}} \times V_{\text{ant}}}{V_{\text{int}}}. \tag{5}$$

In equation [5], Q_{int} represents the amount of charge capacity released by the insulator experimental circuit, which is 0.000302 C. V_{int} represents the maximum amplitude of the voltage time-domain waveform measured by the Rogowski coil, which is 13 400 V. V_{ant} represents the maximum amplitude of the voltage time-domain waveform measured by the antenna at 13 m from the insulator, which is 0.65 mV. The final calculation result is $Q_{\text{ant}} = 3.11$ pC. At a distance of 13 m, we can detect a charge capacity of 3.11 pC. To determine the minimum charge capacity, it is essential to build an experimental platform that can control the charge capacity. This will allow for further experimentation in future work to obtain accurate results.

TABLE II
COMPARISON WITH OTHER WORKS

Ref. No	Size	Detection distance
[7]	2×4 m (reception surface)	320m
[8]	Diameter: 360 mm Height: 2284 mm	260m
[10]	100mm×100mm×1.6mm	5m
This work	130mm×130mm×1.6mm with 150mm×150mm×1.6mm	13m

Finally, we compare the antennas discussed in the cited literature with the designed antenna, and the results of this comparison are presented in Table II. Notably, the designed FSHU antenna exhibits an optimal detection distance of 13 m, surpassing that of the other antennas. The designed antenna is small in size and portable in comparison to larger detection antennas.

V. CONCLUSION

In this article, an FSS operating in the corona discharge band is designed to enhance the sensitivity of corona discharge detection antenna. This FSS comprises two patches: the upper patch consists of twisted cross cells, and the bottom patch consists of cross cells loaded with inductive elements. Simulation software is then used to model and compute the FSS, resulting in the S_{11} curve. The designed FSS exhibits an $S_{11} < -10$ dB within the frequency range of 100–500 MHz. Each cell is 10 × 10 × 1.6 mm. A total of 15 × 15 cells of the FSS are loaded onto the final designed detection antenna. After the simulation calculations, we determine that when the distance between the FSS and the detection antenna is 20 mm, the antenna exhibits the lowest S_{11} curve and the strongest ability to capture electromagnetic waves. Following that, a corona discharge detection platform is built to evaluate the detection capabilities of the SHU and FSHU antennas. At detection distances of 1, 4, 7, and 10 m, the amplitude of the FSHU antenna is 128%, 130%, 132%, and 135% higher than the SHU antenna, respectively. The FSHU antenna is still able to detect corona discharge signals at a detection distance of 13 m.

Our designed antenna still faces several challenges. First, the gain of the antenna presented in this article requires enhancement. Future gains can be improved by optimizing the microstrip layer's shape. Second, the dielectric substrate material used for the antenna is FR4, which contributes to an increase in the antenna's weight. Low-density materials such as polyethylene terephthalate (PET) could be used in future designs to reduce the antenna's weight.

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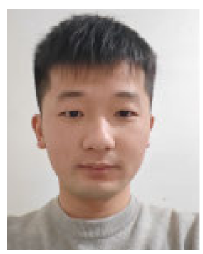

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